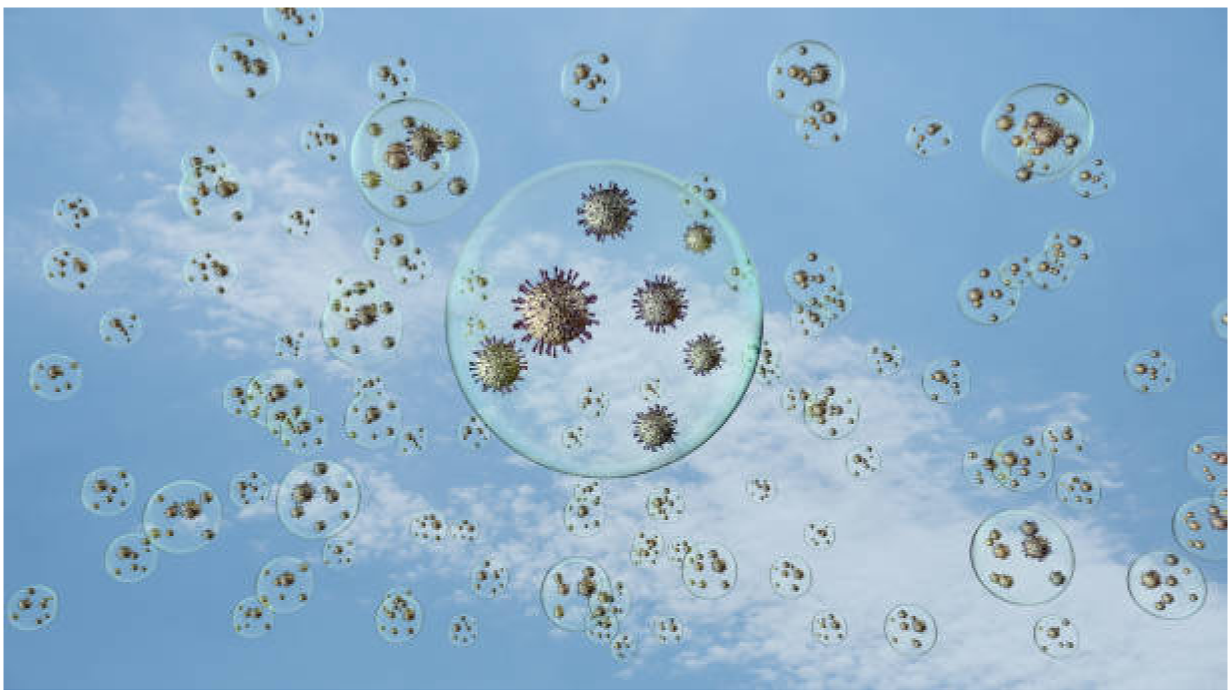
Sampling and evaluation of airborne microflora

CH5200 - Food Microbiology
Experiment 2b



Group 2		
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1 General Introduction

Compared to soil, food, or even water, air is a harsh and hostile environment to microbial growth due to the effects of radiation and desiccation (Adams et al., 2024). However, microbes are still able to disperse through airborne routes, contaminating any suitable, exposed surface for growth. In this experiment, different air sampling instruments were utilized and compared to evaluate the microbiological status of public airspaces, assessing the risk presented by airborne microflora as well as evaluating the performance of the different instruments.

2 Materials and Methods

In this experiment, three air sampling instruments were used: MAS-100 NT® and Hycon RCS Plus, with each apparatus sampling 100 L of air, and Emtek P100 sampling 200 L of air. Sampling was conducted in Classroom 1 (CR1) lecture theater on Level 1 in the School of Biological Sciences at NTU. The airspace had a noticeably musty odor and was an enclosed area with little ventilation. The agar plates/ strips from the various air sampling instruments were then incubated until a suitable amount of growth was observed, and the number of colonies was examined. Further isolation and characterization of pure bacterial colonies (excluding fungi) were done, and an evaluation of the bacterial diversity was made.

3 Results and Discussion

Air sampling was performed inside Classroom 1 (CR1) of the SBS Building using various instruments, namely the Emtek P100, MAS, and Hycon. The colony count and concentration recorded from the air sampling are recorded in Table 1.

Table 1: Colony count and microorganism concentration of CR1 located indoors within the SBS Building, determined using Emtek P100, MAS and Hycon air samplers.

Air sampling method (Instrument used)	Colony count/ $\log_{10}$CFU	Concentration of microorganisms/ $\log_{10}$CFU/m^3
Emtek P100 (Group 9)	16	1.9
MAS (Group 4)	14	2.1
Hycon (Group 2)	Not countable	Not countable

From Table 1, it can be inferred that the MAS instrument yielded a higher concentration of microorganisms in the air sampled, with Emtek P100 yielding a slightly lower concentration (the difference is insignificant, <0.5 $\log_{10}$CFU/m^3 difference). On the other hand, Hycon did not yield a countable result due to condensation smear on the agar strip (Annex Figure A2), resulting in a sample with little/ no distinct colonies.

Colonies on the agar plates/ strips were then streaked on separate agar plates for further identification. The identified bacterial groups are recorded in Table 2.

Table 2: Microorganism species identified in CR1 SBS Building using Emtek P100, MAS and Hycon air samplers.

<table>
<tr><th>Emtek P100 (Group 9)</th><th>MAS (Group 4)</th><th>Hycon (Group 2)</th></tr>
<tr><td colspan="3">Bacillus spp.</td></tr>
<tr><td colspan="2">Enterobacteriaceae spp.</td><td>-</td></tr>
<tr><td>Staphylococcus spp.</td><td>-</td><td>Staphylococcus spp.</td></tr>
<tr><td>-</td><td>Pseudomonas spp.</td><td>-</td></tr>
<tr><td colspan="3">Fungi</td></tr>
</table>

The subdivision of microorganism in groups is located in the annex (Annex Table B1)

3.1. Common microorganisms found in air

In CR1 (our sampling location), the presence of *Bacillus spp.* and fungi were detected by all three instruments. The air is mainly dominated by Gram-positive rod-shaped bacteria (e.g., *Bacillus spp.*). Their thick cell wall structure—mainly composed of peptidoglycan—acts as a protective barrier, allowing for increased viability. Additionally, both *Bacillus spp.* and fungi are capable of producing spores that can survive in the air, further supporting their survivability and persistence in the air. *Enterobacteriaceae spp.* and *Staphylococcus spp.* were also detected and attributed to human origin, as CR1 is a lecture theater where human occupancy is common. Another contributor to the diversity of the airborne microflora in CR1 is the carpet flooring, which is not frequently cleaned and thus may accumulate organic waste, including dirt and dust from the constant foot traffic.

Comparing with the species analysis of other locations (Annex Table B1), similar microbial diversity (excluding fungi, as the experiment excludes fungal identification) was observed, with *Bacillus* spp. and fungi being ubiquitous and other strains including *Pseudomonas spp.*, *Staphylococcus spp.,* and *Enterobacteriaceae spp.* also being present. Noticeably, *Lactobacillus spp.* were found in the Clade Room, which is a probiotic that is commonly traced to human origin, which aligns with the Clade Room being a communal lounge that hosts students frequently (Liu et al., 2022).

3.2. Comparison between indoor and outdoor air quality

Compared with the microbial yield of other locations (Annex Table A1), both indoor and outdoor environments have relatively similar microbial loads (i.e., <0.5 $\log_{10}$ CFU/m^3 difference). While microbial loads show no major differences, the difference in microflora diversity was distinct (focusing on bacteria diversity, excluding fungi), with Gram-negative bacteria observed mostly in indoor environments due to increased susceptibility to outdoor environmental stressors like desiccation and UV radiation. In particular, *Staphylococcus spp.* were only observed in indoor environments. *Staphylococcus spp.* are commonly found on human skin, in the nose, and in mucous membranes. Thus, their presence is attributed to a heightened human presence in indoor locations.

A relatively unexpected result was the presence of *Enterobacteriaceae spp.* in the air sampled at SBS Building Level 2 outdoor area. *Enterobacteriaceae spp.* are Gram-negative bacteria that are susceptible to desiccation, making them less common in open-air environments, albeit not completely impossible. The detection of *Enterobacteriaceae spp.* could be attributed to the abundance of vegetation and soil in the surrounding area of the outdoor area, or the presence of animal (including birds and rodents) waste, all of which are potential sources of *Enterobacteriaceae spp.*

3.3. Location with the highest yield

The highest yield (2.6 $\log_{10}$CFU/m^3) was unexpectedly found at LKC Bridge, which is an open-air environment. This may be due to a recently produced aerosol from a human or animal source (Adams et al., 2024). Another possibility is human error or random fluctuation. The high yield cannot be validated as sampling was only conducted by one group, without additional replicate results. In general, the microbial load across the different locations ranged from 1.8 to 2.6 $\log_{10}$CFU/m^3, which is within the WHO's recommended limit for airborne microbial levels at 3.0 $\log_{10}$CFU/m^3 (Heseltine & Rosen, 2009).

3.4. Comparison of the three instruments

The Emtek P100 uses a vacuum pump attached with a remote head for direct impaction onto an agar plate, highlighting its suitability for high-grade cleanroom settings that require remote air sampling. MAS operates through direct impaction on an agar plate via a perforated head and is well suited for routine monitoring in controlled environments. Hycon uses centrifugal impaction onto an agar strip and is best suited for general environmental monitoring where simplicity, portability, and ease of use are preferred over advanced configurability.

While the microbial load results of all groups (Annex Table A1) does indicate a slight difference in yield, it cannot be concluded that any instrument has a higher yield compared to the others, as the differences observed were within 0.5 $\log_{10}$CFU/m^3 of each other (results are only significant at >0.5 $\log_{10}$CFU/m^3 difference). Additionally, no replicates were performed, hence variations in microbial yield may be associated with random environmental fluctuations. Hence, it can be concluded that the three instruments were of similar effectiveness and accuracy compared to each other. However, this should be further validated by an additional measurement involving sufficient replicates.

4 Conclusions and Recommendations

In conclusion, an evaluation of the air quality in various locations revealed that the majority of the airborne microflora is made up of fungi and *Bacillus spp.*, with Gram-negative bacteria being more prevalent indoors. The microbial loads across the different locations were assessed to be of acceptable quality according to WHO standards and thus assessed to be of acceptable risk (Heseltine & Rosen, 2009).

The performances of the three selected instruments did not show significant differences with each other, which suggests that all three are acceptable for use without any major biases. However, the Hycon was noticeably more difficult to interpret, as the agar strips were prone to condensation smear, which happened to two groups out of four. Hence, extra caution should be exercised when using the Hycon. Furthermore, replicate measurements should be performed to reduce the impact of random fluctuation.

Emtek P100, MAS, and Hycon are all recognized tools for microbial air monitoring, and selecting the most appropriate instrument should be guided by environmental requirements, regulatory considerations, and operational workflow.

References

Adams, M. R., McClure, P. J., & Moss, M. O. (2024). *Food microbiology*. Royal society of chemistry.

Heseltine, E., & Rosen, J. (Eds.). (2009). WHO guidelines for indoor air quality: dampness and mould.

Liu, M., Gan, Z., Shen, B., Liu, L., Zeng, W., Li, Q., & Liu, H. (2022). Occupants contribute to pathogens and probiotics in indoor environments. *Building and Environment, 213*, 108863.

Annex

Table A1: Colony count and microorganism concentration of the four chosen locations (comparison across all groups) located within the SBS Building, determined using Emtek P100, MAS and Hycon air samplers

Location	**Emtek P100/ log_{10}CFU/m^3** (Flow rate: 200 L/min)	**MAS/ log_{10}CFU/m^3** (Flow rate: 100 L/min)	**Hycon/ log_{10}CFU/m^3** (Flow rate: 100 L/min)	**Average log_{10}CFU/m^3**
Groups 1, 5, 7: Open Air Level 2 SBS Building (Outdoor)	CFU count: 26 $\log_{10}(26/0.2)$ = **2.1**	CFU count: 17 $\log_{10}(17/0.1)$ = **2.2**	CFU count: 7 $\log_{10}(7/0.1)$ = **1.8**	$\frac{2.1 + 2.2 + 1.8}{3}$ = **2.0**
Group 3: LKC Bridge (Outdoor)	Not done	CFU count: 44 $\log_{10}(44/0.1)$ = **2.6**	Not countable	**2.6**
Groups 2, 4, 9: CR1 SBS Building (Indoor)	CFU count: 16 $\log_{10}(16/0.2)$ = **1.9**	CFU count: 14 $\log_{10}(14/0.1)$ = **2.1**	Not countable	$\frac{1.9 + 2.1}{2}$ = **2.0**
Groups 6, 8, 10: Clade room SBS Building (Indoor)	CFU count: 35 $\log_{10}(35/0.2)$ = **2.2**	CFU count: 9 $\log_{10}(9/0.1)$ = **2.0**	CFU count: 15 $\log_{10}(15/0.1)$ = **2.2**	$\frac{2.2 + 2.0 + 2.2}{3}$ = **2.1**

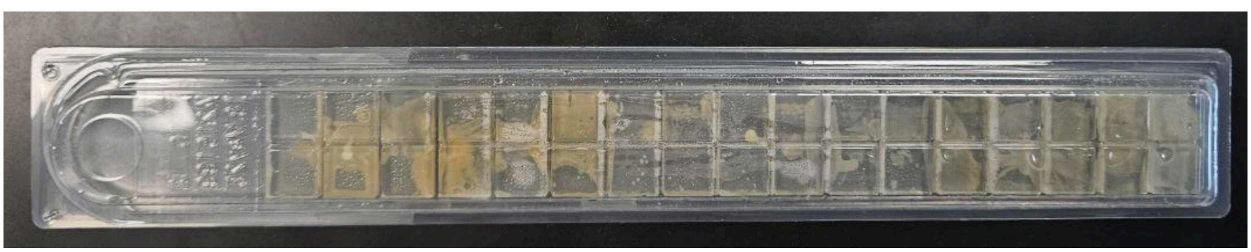


Figure A2: Image of air sample colonies on agar strip used for Hycon instrument (CR1 SBS Building), indicating the presence of condensation smear resulting in non-countable results

Table B1: Microorganism species identified in the four chosen locations (comparison across all groups) located within the SBS Building, determined using Emtek P100, MAS and Hycon air samplers

<table>
<tr><th>Location</th><th>Indoor/ Outdoor</th><th>Emtek P100</th><th>MAS</th><th>Hycon</th></tr>
<tr><td rowspan="3">Groups 1, 5, 7:
Open Air Level 2
SBS Building</td><td rowspan="3">Outdoor</td><td colspan="3">Bacillus spp.</td></tr>
<tr><td>-</td><td>Enterobacteriaceae spp.</td><td>-</td></tr>
<tr><td>Fungi</td><td colspan="2">-</td></tr>
<tr><td rowspan="2">Group 3:
LKC Bridge</td><td rowspan="2">Outdoor</td><td>-</td><td>Bacillus spp.</td><td>-</td></tr>
<tr><td>-</td><td>Other unknown species present</td><td>-</td></tr>
<tr><td rowspan="5">Groups 2, 4, 9:
CR1
SBS Building</td><td rowspan="5">Indoor</td><td colspan="3">Bacillus spp.</td></tr>
<tr><td colspan="2">Enterobacteriaceae spp.</td><td>-</td></tr>
<tr><td>Staphylococcus spp.</td><td>-</td><td>Staphylococcus spp.</td></tr>
<tr><td>-</td><td>Pseudomonas spp.</td><td>-</td></tr>
<tr><td colspan="3">Fungi</td></tr>
<tr><td rowspan="5">Groups 6, 8, 10:
Clade Room
SBS Building</td><td rowspan="5">Indoor</td><td colspan="3">Bacillus spp.</td></tr>
<tr><td colspan="2">Enterobacteriaceae spp.</td><td>-</td></tr>
<tr><td colspan="3">Staphylococcus spp.</td></tr>
<tr><td>-</td><td colspan="2">Lactobacillus spp.</td></tr>
<tr><td>Fungi and Yeast</td><td colspan="2">-</td></tr>
</table>

Table B2: Image of four colonies sampled and isolated from the agar strip used for Hycon instrument (CR1 SBS Building), streaked and cultured on NA (nutrient agar) plates. The colonies were viewed under phase contrast at 1000x magnification (using 100x objective lens with immersion oil)

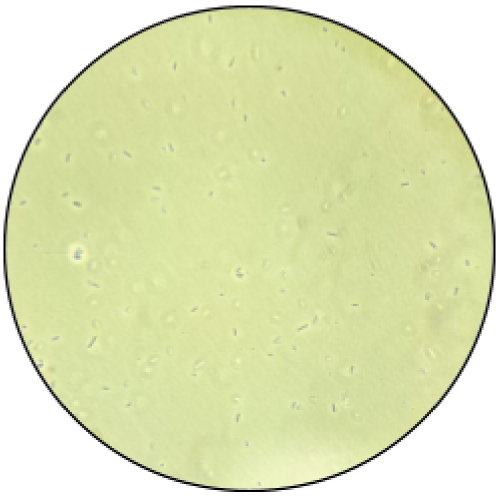
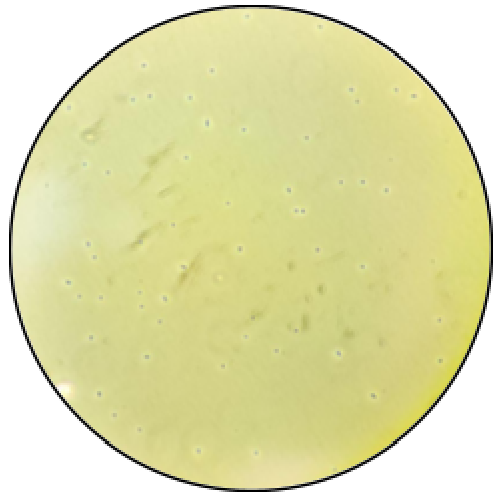
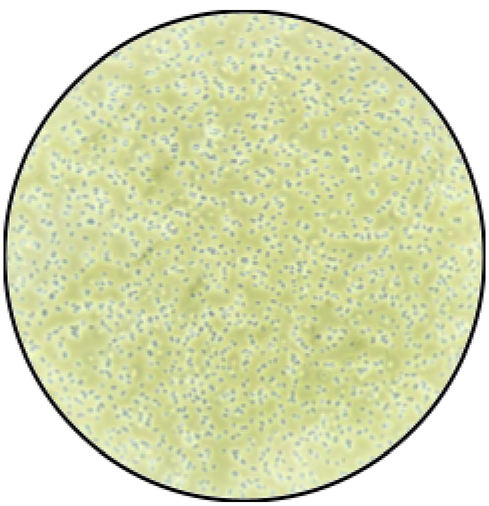
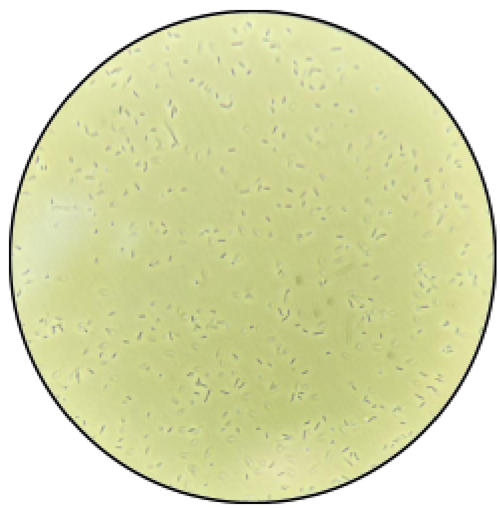
Colony A	Colony B
	
Colony C	**Colony D**
	

Table B3: Identification of the four colonies sampled and isolated from the agar strip used for Hycon instrument (CR1 SBS Building)

Colony	Shape	Motility	KOH	Amino peptidase	Oxidase	Catalase	Group
A	Rod	Motile	- (Gram positive)	- (Gram positive)	-	+	*Bacillus spp.*
B	Coccus	Non motile	- (Gram positive)	- (Gram positive)	-	+	*Staphylo-coccus spp.*
C	Coccus	Non motile	- (Gram positive)	- (Gram positive)	-	+	*Staphylo-coccus spp.*
D	Rod	Motile	- (Gram positive)	- (Gram positive)	-	+	*Bacillus spp.*

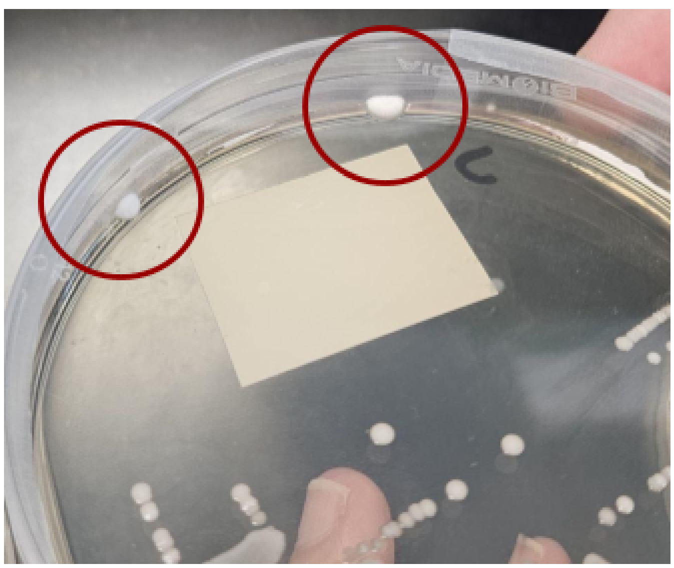


Figure B4: Image of mould colonies found on the NA plate of Colony C, indicating the presence of mould in CR1 SBS Building (using Hycon instrument)